

Mid-term Exam (Graph Neural Networks - Fall 2026)

Full Name:

Student ID:

Note: Students should write in English for evaluation.

1. (Node2vec) Consider an undirected graph G given in the following figure. A random walk has just moved from node A to node B , with $p = 2$ and $q = 0.5$.

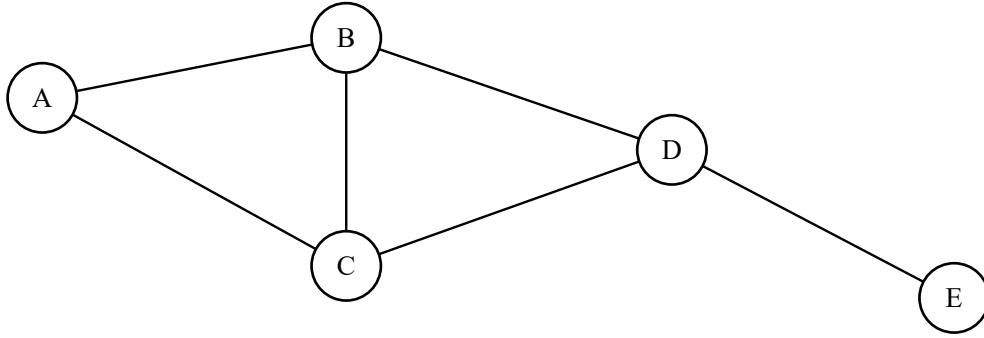


Figure 1. Graph G

The unnormalized transition probability from B to a next node x is $\pi_{Bx} = \alpha_{pq}(A, x)$, where

$$\alpha_{pq}(A, x) = 1/p \text{ if } d(A, x) = 0; \quad 1 \text{ if } d(A, x) = 1; \quad 1/q \text{ if } d(A, x) = 2$$

and $d(A, x)$ is the shortest path distance between node A and node x .

- Explain BFS and DFS, and explain which one a small q makes the walk behave like.
- List all possible next nodes from node B and calculate the unnormalized transition probabilities π_{Bx} .
- Calculate the normalized transition probabilities.

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2. (LINE) Consider an undirected graph G with node embeddings

$$u_A = 1, u_B = 0.5, u_C = -0.5, u_D = -1$$

and context embeddings

$$u'_A = 0.5, u'_B = 1, u'_C = -1, u'_D = -0.5.$$

The second-order proximity is defined as

$$p_2(j|x) = \exp(u'_j u_x) / \sum_k \exp(u'_k u_x).$$

Given $e^{0.25} = 1.28$, $e^{0.5} = 1.65$, $e^{-0.25} = 0.78$, $e^{-0.5} = 0.61$.

- a) What is second-order proximity, and how does it differ from first-order proximity?
- b) Calculate $p_2(A|B)$, $p_2(B|B)$, $p_2(C|B)$ and $p_2(D|B)$.

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3. (GCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0 \\ 3 \end{bmatrix}$$

Assume that the hidden layer of a GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}),$$

where $H^{(k)}$ denotes the output at layer k , $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain how information is propagated in a GCN layer.
- b) Calculate $\tilde{A}$.
- c) Calculate the output of the GCN model at layer $k = 1$.

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4. (GraphSAGE) Consider an undirected graph G of five nodes A, B, C, D and E given in the following figure. Each node has initial features that are the numbers standing next to it (i.e. the initial feature of node 'A' is $h_A^{(0)} = 0.4$). According to the GraphSAGE model with an AGGREGATE that is a MEAN function, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{AGGREGATE}(\{ h_u^{(k-1)}, \square u \in N(i) \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation, $\text{ReLU}(x) = \max(0, x)$, and $N(i)$ is the set of neighbour nodes of node i .

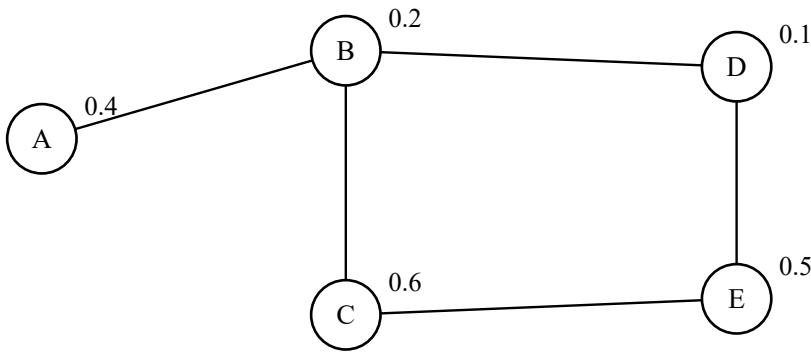


Figure 2. Graph G

- a) What is the main idea behind the GraphSAGE model, and how does it differ from traditional GCNs?
- b) Calculate the feature of each node at $k = 1$.
- c) Calculate a graph-level embedding h_G by using a 'Mean' global pooling when $k = 1$.

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5. (ClusterGCN) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two clusters G_1 and G_2 . Each node has initial features that are the numbers standing next to it. According to the ClusterGCN model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \square u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

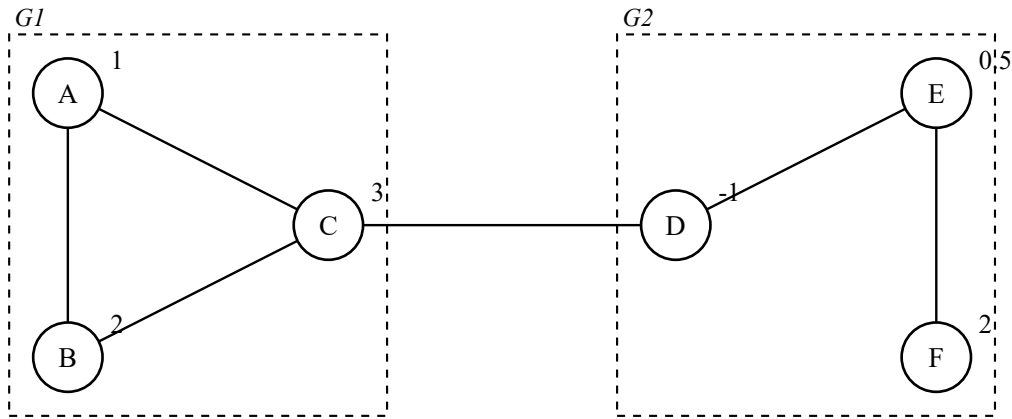


Figure 3. Graph G with two clusters

- What is mini-batch training, and why does ClusterGCN need it?
- Calculate the output representations of all nodes at layer $k = 1$.

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6. (GraphSAINT) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G has a sampled subgraph G_1 . Each node has initial features that are the numbers standing next to it. According to the GraphSAINT model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \square u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

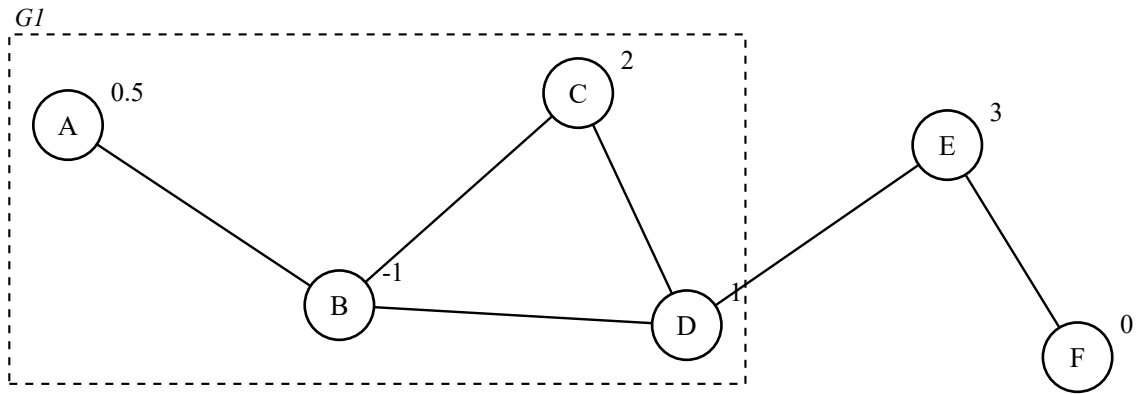


Figure 4. Graph G with a sampled subgraph G_1

- How does GraphSAINT's sampling strategy differ from node-wise or layer-wise sampling methods like GraphSAGE and FastGCN?
- Calculate the output representations of nodes A, B, C and D at layer $k = 1$.

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7. (JK Network) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

Assume that the output of a JK network model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \max(\sigma(\tilde{A} \cdot H^{(0)}), \sigma(\tilde{A} \cdot H^{(1)}), \dots, \sigma(\tilde{A} \cdot H^{(k-1)}))$$

where $H^{(k)}$ denotes the output at layer k , $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain how Jumping Knowledge Networks help mitigate the over-smoothing problem in deep GCNs.
- b) Calculate $\tilde{A}$.
- c) Calculate the output representations at layer $k = 2$.

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8. (GCNII) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \\ -3 \end{bmatrix}$$

Assume that the output of a GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma[((1 - \beta) I_n) \cdot ((1 - \alpha) \tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)})]$$

where $H^{(k)}$ denotes the output at layer k , $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), I_n is the identity matrix, $\alpha = \beta = 0.5$, and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) What are the two major components introduced in GCNII to overcome the depth limitation problem in GCNs?
- b) Calculate $\tilde{A}$.
- c) Calculate the output representations at layer $k = 1$.

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9. (DeepGCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ -2 \\ 1 \\ 0 \end{bmatrix}$$

Assume that the output of a DeepGCN model with a residual connection of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}) + H^{(k-1)}$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain why a residual connection allows a GCN to be made deeper.
- b) Calculate the output representations at layer $k = 1$ and $k = 2$.
- c) Compare $H^{(1)}$ with the output of a plain GCN without the residual connection.